

Grade 8
Unit 5
Lesson 4 Practice

Name _____
Date _____
Period _____

1. The Triangle Inequality Theorem states that the

sum
difference

 of the lengths of any two sides of a triangle must be

less than
greater than

 the length of the third side.

For questions 2–5, determine if each set of side lengths form a triangle.

2. 92 cm, 59 cm, 24 cm

$$92 + 59 = 151 > 24$$

$$92 + 24 = 116 > 59$$

$$59 + 24 = 83 < 92$$

No

3. 5 ft, 5 ft, 10 ft

$$5 + 10 = 15 > 5$$

$$5 + 10 = 15 > 5$$

$$5 + 5 = 10 = 10$$

No

4. 6 in, 8 in, 9 in

$$6 + 8 = 14 > 9$$

$$6 + 9 = 15 > 8$$

$$8 + 9 = 17 > 6$$

Yes

5. 2.2 yd, 3.5 yd, 4 yd

$$2.2 + 3.5 = 5.7 > 4$$

$$2.2 + 4 = 6.2 > 3.5$$

$$3.5 + 4 = 7.5 > 2.2$$

Yes

6. The Converse of the Pythagorean Theorem states that if the lengths a , b , and c of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$, then the triangle is a

- right triangle.
acute triangle.
obtuse triangle.

For each of the following, determine if each set of side lengths form a triangle. If the side lengths form a triangle, determine if it is a right triangle using the Converse of the Pythagorean theorem.

7. $a = 16, b = 30, c = 35$

$$16 + 30 = 46 > 35$$

$$16 + 35 = 51 > 30$$

$$30 + 35 = 65 > 16$$

Triangle

$$a^2 + b^2 = c^2$$

$$16^2 + 30^2 \stackrel{?}{=} 35^2$$

$$256 + 900 \stackrel{?}{=} 1225$$

$$1156 \neq 1225$$

Not a Right Triangle

8. $a = 12, b = 35, c = 37$

$$12 + 35 = 47 > 37$$

$$12 + 37 = 49 > 35$$

$$35 + 37 = 72 > 12$$

Triangle

$$12^2 + 35^2 \stackrel{?}{=} 37^2$$

$$144 + 1225 \stackrel{?}{=} 1369$$

$$1369 = 1369$$

Is a Right Triangle

9. $a = 6, b = 13, c = 19$

$$6 + 13 = 19$$

$$6 + 19 = 25 > 13 \rightarrow$$

does not form a triangle

$$13 + 19 = 32 > 6$$

10. $a = 7.5, b = 10, c = 12.5$

$$7.5 + 10 = 17.5 > 12.5$$

$$7.5 + 12.5 = 20 > 10$$

$$10 + 12.5 = 22.5 > 7.5$$

Triangle

$$a^2 + b^2 = c^2$$

$$7.5^2 + 10^2 \stackrel{?}{=} 12.5^2$$

$$56.25 + 100 \stackrel{?}{=} 156.25$$

$$156.25 = 156.25$$

Is a Right Triangle